



School of Future Tech

Case Study Report

Riddle Challenge Portal (Logic Puzzle Platform)

Your Case Study Title

by

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Figma Link:

<https://www.figma.com/design/oQ81RNztoVnNSiigQNgUWX/Home?node-id=0-1&t=3nYJEMMPK0GLIP2e-1>

PPT Link:

https://docs.google.com/presentation/d/1q9LwHYiLMtVk7D05MazK91FaLGHaPTwy_8n7oIZ2MGY/edit?usp=sharing

Github Repository Link:

<https://github.com/sejaltech/Final-project-of-HTML-CSS-JS.git>

1. Introduction to the Case Study

Puzzle and riddle games are a fun way to improve logical thinking and problem-solving skills. In today's digital world, interactive web applications allow users to enjoy such activities online with engaging designs and instant feedback.

This case study focuses on designing and implementing a **Riddle Challenge Portal** using web technologies such as **HTML, CSS, and JavaScript**. The system allows users to play riddles, submit answers, receive feedback, track scores, and view results on a leaderboard.

The main objective of this project is to create an **interactive and visually engaging platform** where users can test their knowledge and logical thinking through riddles. The system also demonstrates how front-end technologies can be used to build a structured multi-page web application with dynamic user interaction.

2. Problem Statement / Case Background (Abstract)

Background

Many online puzzle or quiz websites are either overly complex or lack engaging user interfaces. Users often lose interest if the interface is not interactive or visually appealing.

A simple yet well-designed riddle platform can provide an enjoyable experience where users can challenge themselves while tracking their performance.

Abstract

This case study presents the design and implementation of a **Riddle Challenge Portal** built using **HTML, CSS, and JavaScript**. The platform allows users to solve riddles, enter answers, receive hints, track their score, and view final results.

The application includes features such as **a timer system, difficulty levels, hints, scoring, progress tracking, and a leaderboard**. CSS animations and responsive design techniques are used to create an engaging user experience.

The project demonstrates how front-end web technologies can be used to develop an interactive browser-based game without requiring complex backend systems.

3. Case Study Design

The **Riddle Challenge Portal** is designed as a multi-section web application with smooth navigation between pages.

Main Sections of the System

Home Page

The home page introduces the platform and allows users to start the challenge. It displays key information such as number of riddles, difficulty levels, and main features.

Play Page

This is the main game interface where users attempt to solve riddles. It includes:

- Riddle display section
- Answer input field
- Hint button
- Timer countdown
- Score tracking

Leaderboard Page

This page displays the top players based on their scores. It helps create a competitive environment where users can try to achieve higher rankings.

Pricing Page

The pricing page explains the concept of free riddles and premium puzzle sets for advanced users.

Forum Page

The forum allows users to discuss riddles, share answers, and interact with other players.

Navigation Flow

Home → Play → Results → Leaderboard → Forum

This flow ensures smooth user navigation and an engaging experience.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Technologies Used

HTML (HyperText Markup Language)

Used to create the structure and layout of the webpages.

CSS (Cascading Style Sheets)

Used for designing the visual appearance of the website, including layout, colors, fonts, and responsiveness.

JavaScript

Used to add interactivity and dynamic behavior to the portal.

Web Development Methods

Modular Page Design

Each page is created separately to maintain a clear and organized website structure.

Responsive Layout

The layout is designed to adjust across different screen sizes.

User Interface Design

The interface is designed to be simple, interactive, and user-friendly.

Navigation System

A navigation menu allows users to move easily between different pages of the portal.

5. Problem Statement / Case Study Implementation Details and Snapshots.

The implementation of the Riddle Challenge Portal is organized using separate files for structure, design, and functionality.

HTML Implementation

HTML files define the structure of each page including:

- Navigation bar
- Buttons
- Content sections
- Tables for leaderboard display
- Interactive sections for riddles

CSS Implementation

CSS files are used to style each page individually while maintaining a consistent design across the portal.

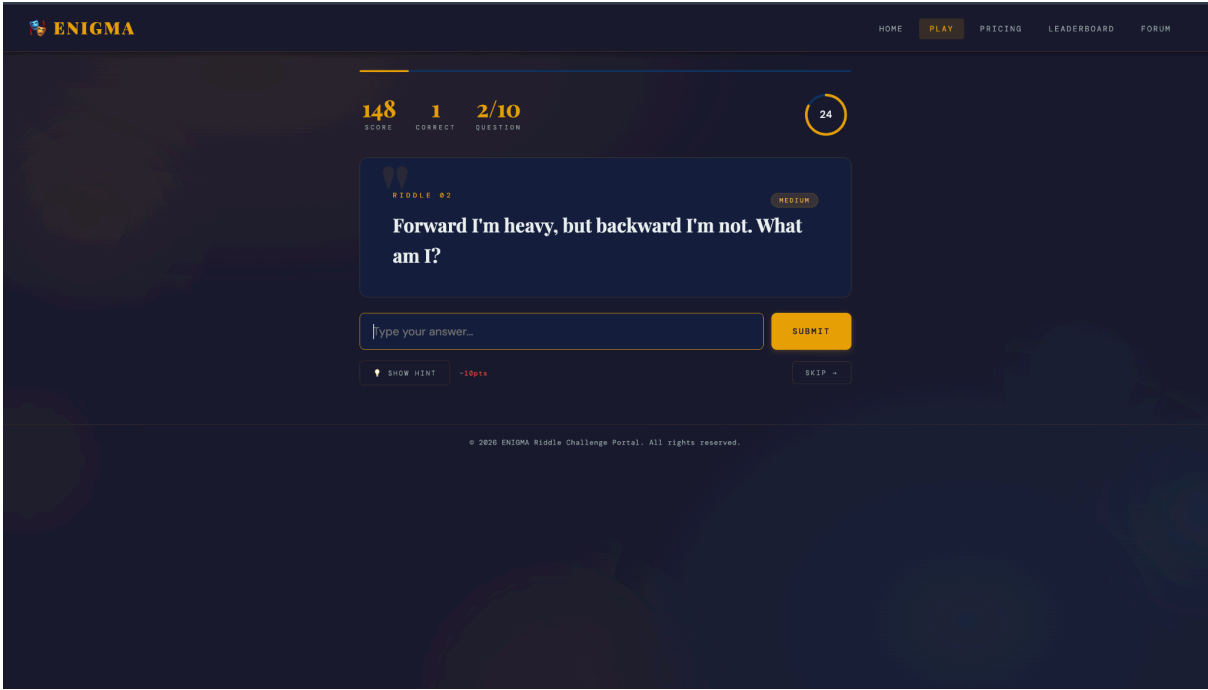
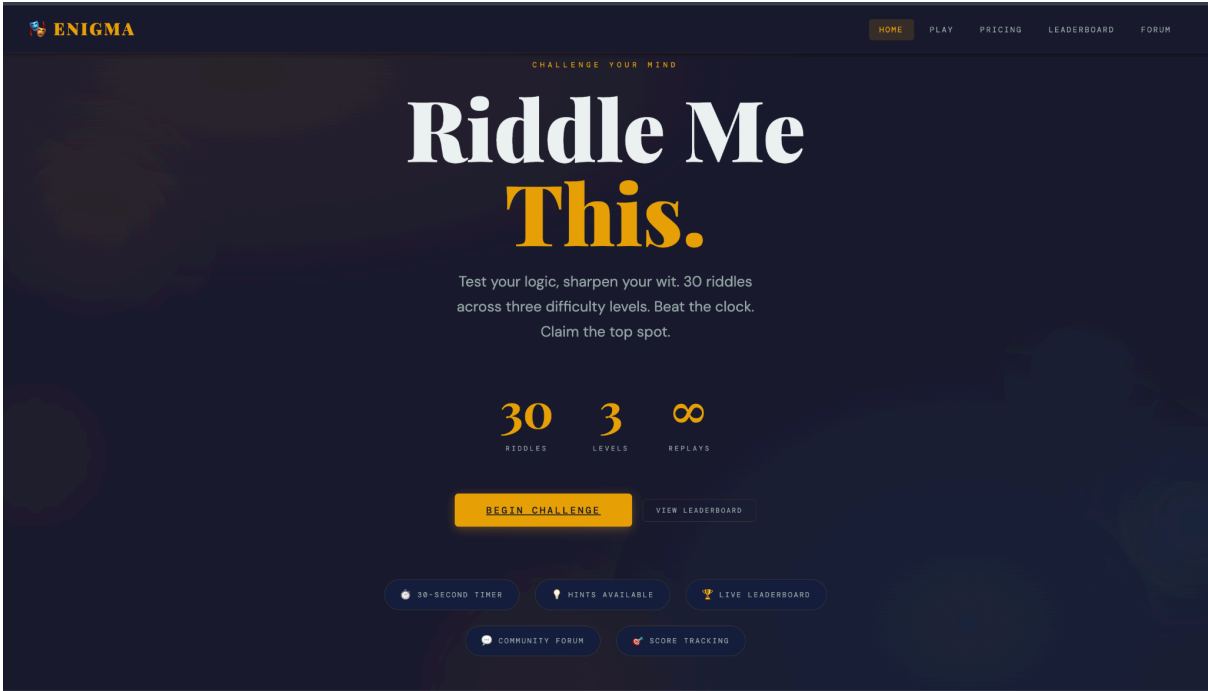
The **base.css** file provides the common styles used across all pages such as fonts, layout structure, and global design elements.

JavaScript Implementation

The **script.js** file handles interactive behavior of the website such as:

- Button interactions
- Page navigation behavior
- Dynamic user interface responses

Snapshots:-



ENIGMA

HOMEPLAYPRICINGLEADERBOARDFORUM

👍

Not Bad!

50% ACCURACY - 5 CORRECT ANSWERS

757

TOTAL SCORE

5/10

CORRECT

50%

ACCURACY

Enter your name for lei.

SAVE SCORE

PLAY AGAIN

VIEW LEADERBOARD

ENIGMA

HOMEPLAYPRICINGLEADERBOARDFORUM

HALL OF FAME

Leaderboard

👤

ajay

757

POINTS

👤

sejall

1252

POINTS

RANK	NAME	CORRECT	ACCURACY	SCORE
1	sejall	8/10	80%	1252
2	ajay	5/10	50%	757

PLAY NOW --

ENIGMA

HOMEPLAYPRICINGLEADERBOARDFORUM

CHOOSE YOUR PLAN

Pricing & Plans

Start free or upgrade for unlimited access to advanced riddles

MONTHLY

YEARLY

SAVE 35%

Starter

Perfect for trying us out

FreeForever

YOUR PLAN

✓ 10 riddles per game

✓ 3 difficulty levels

✓ Leaderboard access

✓ Forum discussions

✗ Advanced riddle sets

✗ Daily challenges

✗ Unlimited hint usage

✗ Priority support

MOST POPULAR

Pro

For serious puzzlers

₹99/month

UPGRADE TO PRO

✓ Everything in Starter

✓ 50+ advanced riddles

✓ Daily challenges

✓ Unlimited hints

✓ Custom difficulty modes

✓ Advanced analytics

✗ Elite riddle collection

✗ Priority support

Elite

Maximum experience

₹999/month

UPGRADE TO ELITE

✓ Everything in Pro

✓ 100+ elite riddles

✓ Weekly tournaments

✓ Custom riddle creation

✓ Multiplayer mode

✓ Extended time modes

✓ 24/7 priority support

✓ Exclusive badges & titles

Frequently Asked Questions

Can I upgrade or downgrade anytime?

+

Is there a free trial?

+

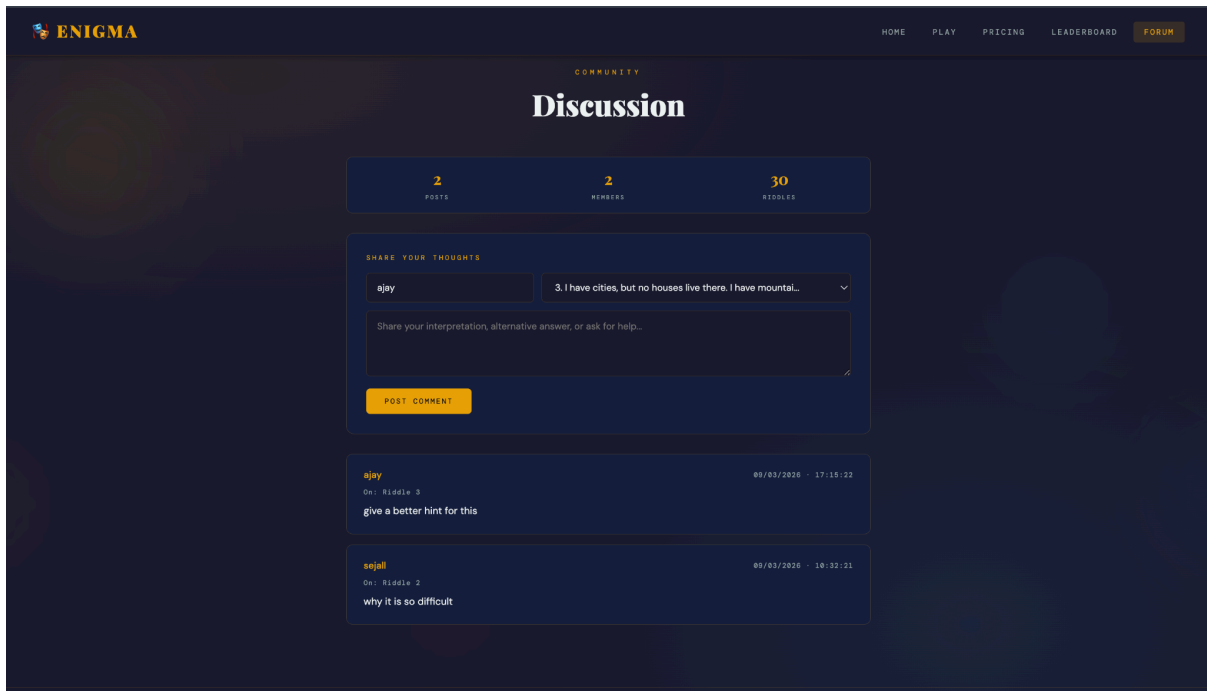
What payment methods do you accept?

+

Can I cancel anytime?

+

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These screenshots demonstrate how the system works and how users interact with the platform.

6. Problem Statement / Case Study Results and Conclusion.

Results

The developed Riddle Challenge Portal successfully provides an interactive platform where users can solve riddles and track their performance.

The use of CSS animations and structured layout improves the visual appeal and makes the application more engaging.

JavaScript logic effectively manages game functionality including answer checking, score calculation, hints, and timer control.

Conclusion

This case study demonstrates how **front-end web technologies** can be used to build an interactive puzzle game application.

The project highlights the importance of combining **good user interface design with logical programming** to create an enjoyable user experience.

The Riddle Challenge Portal can be further improved by adding features such as:

- User accounts
- Database-based leaderboard
- More riddles and difficulty levels
- Mobile optimization

7. References

- HTML Documentation – W3Schools
 - CSS Documentation – MDN Web Docs
 - JavaScript Documentation – MDN Web Docs
 - Google Fonts for typography resources
 - Various online tutorials related to web development and interactive UI design
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